# Patrick Thangarajah

FINITE ELEMENT ANALYSIS | COMPUTATIONAL BIOMECHANICS | LS-DYNA | MODEL VALIDATION

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## Summary

Mechanical Engineering MASc graduate with experience in finite element (FE) modelling, FE model validation, and simulation-based design, applied to occupant safety and impact biomechanics.

## Skills

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| **Occupant Safety and Crash Analysis** | LS-DYNA, nonlinear FEA, explicit dynamics, crash simulation, injury biomechanics |
| **Model Development and Validation** | FE model validation, physical test correlation, ISO 18571, material models, contact, boundary conditions |
| **Simulation and Data Analysis** | LS-PrePost, MATLAB, JMP, design of experiments (DOE), Python, Microsoft Excel |
| **CAD and Engineering** | SolidWorks, engineering drawings, technical reports, technical presentations |

## Education

**Master of Applied Science in Mechanical and Mechatronics Engineering** — *Waterloo, ON, Canada*

University of Waterloo — *Sep 2023 - Jul 2026*

- Research in occupant safety and biomechanics

**Bachelor of Applied Science in Mechanical Engineering, Co-op Program** — *Vancouver, BC, Canada*

University of British Columbia — *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option

## Work/Research Experience

**Graduate Research Student - Impact Mechanics and Material Characterization Group** — *Waterloo, ON, Canada*

University of Waterloo — *Sep 2023 - Jul 2026*

- Validated a THOR-5F ATD finite element model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Investigated material response, contact interactions, boundary conditions, and model behavior to understand discrepancies between simulation predictions and experimental measurements
- Designed and analyzed a 104-simulation DOE-based FE parametric study evaluating the effect of seat belt restraint anchor locations on rear seat occupant response, identifying a rearward and inboard position that reduced peak chest deflection by 16% below the study average
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

**Summer Research Student** — *Vancouver, BC, Canada (Remote)*

Centre for Heart Lung Innovation — *May 2022 - Aug 2022*

- Built a Jupyter Notebook transfer-learning script to diagnose diabetic kidney disease using an emerging imaging platform
- Improved diagnostic accuracy by 10% through model optimization, training refinement, and comparison with existing tissue-slide images

**Mechanical Engineering Intern** — *Airdrie, AB, Canada*

GN Corporation — *Jan 2022 - Apr 2022*

- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications
- Reduced fiberglass cutting waste by 50% through process analysis, and proposed a material handling workflow projected to halve pick-and-place time

## Projects

**Simulation of Cervical Fractures in American Football (ME 720 Course Project)** — *Waterloo, ON, Canada*

University of Waterloo — *Sep 2023 - Dec 2023*

- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

**Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project)** — *Waterloo, ON, Canada*

University of Waterloo — *Jan 2024 - Apr 2024*

- Verified 5 polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all 5 foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

## Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.